

Offline Phase



Legal Documents



Document Parsing



LLM Entity &
Relation Extraction



Incremental
KG Builder



Chapter &
Article Summaries



Knowledge Representations

Document Forest · Knowledge Graph · Ontology



Online Phase



User Query



Stage 0: Query Analyzer

Intent + Type + Abbreviation + Entity

Knowledge Graph

Tier 1: Structural



LLM-Guided
Tree Traversal

Forest → Domain → Doc
Doc → Chapter → Article

Document Forest

Tier 2: Semantic



Dual-Level
Retriever

Low: keyphrase + semantic + PPR + concept
High: theme + hierarchy expansion

Tier 3: Fusion

Knowledge Graph



Semantic Bridge — RRF + KG Expansion

$w_{T1}=1.2, w_{T2}=0.7, w_{KG}=1.0, k=60$



Answer Generation



Legal Answer